

Conflict vs View Serializability

Database Management Systems — Gate Smashers Quick Revision & Exam Guide

1. Foundation & Hierarchy

Conflict Serializability

Uses **Conflict Pairs** and a **Precedence Graph**.

- **No Cycle:** Conflict Serializable ✓
- **Cycle:** Not Conflict Serializable ✗

View Serializability

Uses **View Equivalence** testing.

Establishes serializability even when the precedence graph contains a cycle, specifically in the presence of blind writes.



2. Blind Writes & Core Mechanics

What is a Blind Write?

A **Blind Write** occurs when a transaction performs a $WRITE(A)$ operation without performing a preceding $READ(A)$ on the same data item.

Normal Read-Write:

$R_1(A) \rightarrow W_1(A)$

Blind Write:

$W_1(A)$ (No prior Read)

Lecture Example: Why Blind Writes Matter

Consider an initial state $A = 100$ with three sequential transactions:

T1: $R(A), W(A) \rightarrow A = A - 40$

T2: $W(A) \rightarrow A = A - 40$

T3: $W(A) \rightarrow A = A - 20$

Evaluation: $100 \rightarrow 60 \rightarrow 20 \rightarrow A = 0$

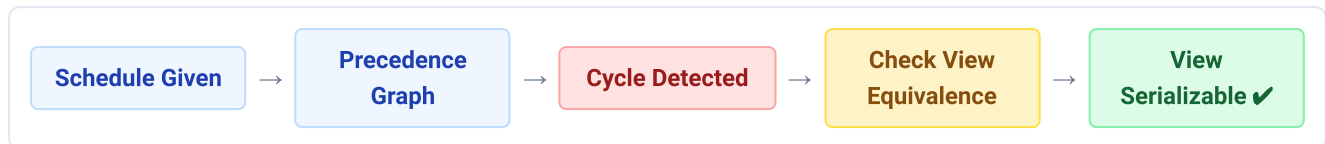
Rearranging blind write operations creates a new schedule that produces a **cycle in the precedence graph** (failing Conflict Serializability), yet preserves the exact final result ($A = 0$), making it **View Serializable**.

3. Resolving Precedence Graph Cycles

⚠ CRITICAL EXAM RULE

A Cycle in Precedence Graph does NOT mean "Not Serializable"!

It strictly means: **Not Conflict Serializable**. You must immediately check for View Serializability before making a final determination.

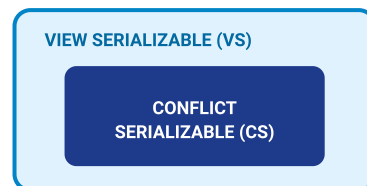


4. Theoretical Relationship & Venn Model

Conflict Serializable $\Rightarrow$ View Serializable

View Serializable $\nRightarrow$ Conflict Serializable

Set Notation: $CS \subseteq VS$



5. Quick Feature Comparison

Feature	Conflict Serializability	View Serializability
Primary Technique	Precedence Graph / Conflict Pairs	View Equivalence Testing
Cycle Detection	Main Test (Cycle = Fails)	Not directly checked via graph
Blind Writes	Restricted / Cannot handle cycles	Handles blind write schedule re-ordering
Scope & Strictness	Strict / Smaller Class (CS)	Broad / Larger Class (VS)

6. Exam Strategy & True/False Traps

✓ **Statement 1:** "If precedence graph has a cycle, schedule is not conflict serializable." — **TRUE**

✗ **Statement 2:** "If precedence graph has a cycle, schedule is not view serializable." — **FALSE**

✓ **Statement 3:** "Every conflict-serializable schedule is view-serializable." — **TRUE**

✗ **Statement 4:** "Every view-serializable schedule is conflict-serializable." — **FALSE**

📌 **One-Line Exam Memory:** "C is inside V. Graph cycle eliminates CS, but VS may still succeed via Blind Writes."